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Work History

Overall description of my research career:

My general research interests are (1) Develop novel XXXX techniques (hardware) for recording biological samples that require ultra-high sensitivity and resolution. Meanwhile I develop computer algorithms (software) to automatically process and analyze the XXXX (data) taken by these advanced techniques. (2) Investigate important biophysical perspectives of the XXXX (normally require the most advanced XXXX techniques developed either by myself or others, since the old techniques do not allow answering the questions I am studying.) These important perspectives span from the most microscopic properties of individual protein molecules and their interactions inside a cell to the connectivity of the most complicated macroscopic network between cells, the neuronal circuits of the mammalian brain.

In particular, since my Ph.D., I have been investigating molecular mechanisms of the XXXX, which plays critical roles in developing and maintaining normal functions of all cells -- especially XXXX. Abnormality of XXXX causes multiple neuronal degenerative diseases, such as XXXX. Exploring this largely unknown field is a key step towards understanding and curing these diseases. From my postdoctoral era, I am also investigating how XXXX. These studies will greatly expand our current knowledge about how a XXXX brain work. Important scientific progress often follows advance of novel technology. As all my researches are at the most frontier of their fields, there are generally no adequate assays to study these outstanding questions. To encounter such obstacles, I invented novel tools, which greatly helped not only answering my questions of interest but also opened new windows for the related research communities.

* As of XX/XX/XXXX, Google Scholar shows total of XXX citations for my XX papers. Recognizing my achievements in the field, I have been invited to review XX journal papers and XX book for different publishers.

XXXX background:

In the cell small molecules, such as gases and glucose, diffuse to where they are needed. Large molecules synthesized in the cell body, intracellular components such as vesicles, and organelles such as mitochondria are too large (and the cytosol too crowded) to diffuse to their destinations. Motor proteins fulfill the role of transporting large cargo about the cell to their required destinations. XXXX are motor proteins that transport such cargo by walking unidirectionally along microtubule tracks hydrolyzing one molecule of adenosine triphosphate (ATP) at each step. Such intracellular transport system is very similar to the highway transportation system in our human society: the trucks (XXXX motor proteins) burning gas (ATP) deliver cargoes (cellular organelles) from the factories (center of the cell, where these cargoes assemble) to their destinations (periphery and other specific structures of the cell) on the highway system (cytoskeleton system, include microtubule, actin and intermediate filament etc.). It is

obvious that a functional intracellular system plays critical roles for maintaining normal functions of all cells in human body. This is particularly important for those polarized cells (cells having asymmetric shape and function) in all four types of human's primary body tissues (epithelial cells, fibroblast cells, muscle fiber cells and neurons of the epithelial, connective, muscle and nervous tissue, respectively). Intracellular transport is the key to establish and maintain such polarity, since molecule diffusion is uniform in all directions. Abnormality of intracellular transport in these polarized cells causes serious degenerative diseases. The well-known examples are Alzheimer's disease and Parkinson's disease.

One of my main research interests since XXXX is to understand the molecular mechanisms of XXXX motor (the truck) transport and regulation. Since the discovery of XXXX in the early XXXXs, it rapidly draw attention to biologists in all fields. Due to its interesting and important functions to all cells, it became an important model molecule for studies in biochemistry, biophysics, cell biology, structure biology and neuroscience. Through the history of studying this model molecule, vast novel biotechniques, especially those in the biophysics field have been invented and improved. These important biotechniques, including 3D cryo electron tomography and nano-meter resolution single molecule fluorescent microscopy not only helped us understand XXXX motors, but also pushed forward studies of other model molecules such as DNA and RNA. After 20 years of elaborate research, although biochemical and biophysical properties of XXXX-1 have been extensively studied *in vitro* (the purified proteins in test tubes or reaction chambers), its properties and regulatory mechanisms are still barely understood in living cells. There is no doubt that live cell study will lead us to closer understanding of XXXX motor's true properties. And the mechanisms of how XXXX work with other components of the intracellular system can hardly be faithfully reconstructed *in vitro*. It is only when finally gaining enough knowledge about XXXX in living cells that we might understand the mechanisms of diseases caused by intracellular transport dysfunction.

The difficulty of studying XXXX motors in living cells comes in three folds: (1) It is a much more crowded environment inside the cells than *in vitro*, thus "noise" from other molecules hinders detection of the rapid movement of XXXX motors. (2) In cells, the majority of XXXX are kept in an inactive form to reduce unnecessary ATP consumption yet remain a big pool of motors for transport on demand (similar to the engine-shut-down U-Haul trucks in the parking lot). On the other hand, each cellular organelle (cargo) undergoing active transport is moved by only one or a few XXXX motors. Difficulty of singling out the active XXXX signal from a huge inactive pool again hinders the detection of XXXX motors. (3) Due to the much higher noise and lower signal level in living cells, it requires better data processing methods to filter the signal from background noise for analysis. These difficulties hesitate live cell studies for XXXX motors. Even for the best known member XXXX-1 there are many outstanding questions remain: (1) From earlier biochemistry experiments we know that over 95% of the XXXX-1 motors purified from cells do not have ATP hydrolysis activity. XXXX-1 thus thought to be kept inactive by unknown auto-inhibition mechanism to prevent unnecessary energy consumption. However detail about this mechanism is unclear and how the motor transit from the inhibited state to an active state for transport is also unknown. (2) After activation, XXXX-1 motors move along microtubule tracks. Immunocytochemistry and electron microscopy still images show that XXXX-1 motor either attached to microtubule alone or link between microtubule and cargo. The motile properties of XXXX-1 moving along microtubule after activation are unclear in living cells. (3) There is a very complicated network of microtubule similar to our highway system in every cell. XXXX-1 needs to distribute its cargoes to specific locations in a cell. Once being activated how does XXXX-1 motor know which microtubule

track to take is unknown. (4) Once XXXX-1 arrives its destination, how does it unload its cargo and the final fate of the motor is also unknown.

*** MY ORIGINAL CONTRIBUTIONS AND THEIR IMPACT ON THE FIELD ***

Project 1: Year XXXX-XXXX, Ph.D. thesis research supervised by XXXX.

Successfully developed and utilized novel bio-imaging techniques for ultra-sensitive, accurate and quantitative studies of XXXX motor proteins and their interacting cellular components in living cells. This contribution is also general to the biophysics field since these techniques are applied to studies of molecules other than motor proteins.

One of my main Ph.D. research projects is to develop imaging techniques to monitor XXXX in living cells. Combining genetic labeling tricks and ultra-sensitive fluorescence detection optics, I successfully designed a novel imaging system that allows the first time recording fast unidirectional movement of a single motor protein inside a living cell. Analyzing the motility behavior of single XXXX-1 proteins, the speed and processivity of this motor protein was measured and compared to their *in vitro* properties. To rapidly extract overall trafficking information from records of tens of thousands of individual molecules, I developed an image processing algorithm based on analyzing the statistics difference of the fluorescent signal over space and time.

I first developed novel techniques to encounter these three folds of difficulties. In specific, I developed a labeling method utilizing XXXX of fluorescent protein to genetically label a version of constitutive active XXXX motor. By this way I increased the labeling signal several folds and largely reduced the proportion of inactive motors. Modifying a total internal reflection fluorescence microscope, I am able to illuminate only a shallow fraction inside the cell to eliminate the background noise from the out of focus motor proteins or other cellular molecules. By combining these methods of increasing signal and decreasing noise, for the first time one was able to capture the rapid processive movements of XXXX motors at the speed of XXXX frames per second in the living cells (XXXX times faster than the real-time capturing rate of the movie videos). The computer software I developed allowed one to use a simple algorithm to rapidly process the recorded images and reveal overall motor protein motile behavior over a period of time despite the high background inside the living cells.

This project led to a publication in XXXX, one of the most highly rated journals in the XXXX field (XXXX). As of XXXX, XXXX search shows total of XXXX citations. A detailed version of this method is invited to publish in XXXX (XXXX." XXXX; XXXX citation).

Significance and Impact:

(1) Since the first series of *in vitro* single bio-molecule observation being made in the early 1990s, tremendous amount of effort has been made to try to observe single bio-molecule events inside the live cells. However, there was NO success in almost 20 years. Using XXXX as a model protein, I developed the first method, which allows direct observation of XXXX inside the XXXX living cells. It has been highlighted in multiple journal review articles as important progress for XXXX and for XXXX research *in vivo*:

- Professor XXXX (Department of XXXX, University of XXXX) has highlighted my work in his review about XXXX [XXXX] as: “XXXX et al. developed a XXXX and used it to XXXX. This much better defined XXXX provides strong evidence that XXXX, thereby ruling-out the hypothesis that XXXX.”
- Professors XXXX and XXXX (XXXX) has highlighted my work multiple times in their review about the research on XXXX [XXXX]. In this review, my work has been emphasized as “of outstanding interest” for its importance in leading the transition from *in vitro* to *in vivo* research about XXXX. In this review, 3 of my other papers are also highlighted (XXXX) Prof. XXXX should be willing to write me a letter. He also cited my other works in many other reviews and his own research papers.

(2) Being the first direct measurement of XXXX properties in living cells, my results have been cited by many other groups as calibration standard when they develop advanced live cell imaging techniques or study XXXX *in vivo*:

- Dr. XXXX’s group (XXXX) has used the XXXX speed I reported as reference in their novel XXXX technique, XXXX [XXXX]: “As a challenging test case with XXXX, we imaged XXXX. XXXX at a typical speed (ref XXXX, my XXXX paper) of about XXXX, corresponding to XXXX.”
- Professor XXXX’s group (XXXX) and Professor XXXX’s group (XXXX) have used the XXXX run length I reported as reference in their study of XXXX [XXXX] as: “...XXXX (XXXX, XXXX).”

(3) Included in text books: XXXX by XXXX (Chapter XXXX, XXXX)

(4) Invitation for talks: Due to the significance of this work, I was invited to give a platform presentation in the XXXX Annual Meeting, XXXX (XXXX). It is privilege to become one of the handfuls of speakers to give a talk in a symposium, selected from hundreds of abstract applications.

(5) Grants: The results and the scientific importance of this project were central to a successful application of a \$ XXXX M funding support from NIH (XXXX).

Project 2: Year XXXX-XXXX, collaborate research project supervised by XXXX and XXXX.

XXXX have been extensively studied as XXXX in humans and animal. However, how such virus bind to and enter the cell and consequentially traffic to the right compartment of the cell to cause tumor-generating infection is poorly understood. In this project, we studied XXXX infectious mechanism by real-time imaging XXXX and XXXX. I designed and built a XXXX that allows simultaneously imaging XXXX particles and infected XXXX in two separate channels in real time. Such true synchronization allows recording of the rapid and transient interactions between viral particles and cell organelles without any time delay. I also developed image processing software that corrects any spatial distortion between the two recording channels allows identifying co-localization between the two at nanometer resolution. This advanced imaging and computer-aid image processing system is the technique foundation of the project. Combining with molecular and genetic manipulations to the cell, we

identified active transport pathway of XXXX and important cellular factors contributing to the infectious pathway.

This work was published in XXXX, one of the most highly rated journals in the XXXX field (XXXX). As of XXXX, Google Scholar search shows total of XXXX citations.

Significance and Impact:

(1) The effective infection of tumorous virus, XXXX relies on XXXX receptors on the cell membrane. The mechanisms by which XXXX receptors guide intracellular virus transport are poorly characterized. Using live cell real-time imaging, we provide a rationale for transport of XXXX through the XXXX, demonstrate a novel XXXX transport pathway that is regulated by a lipid, and implicate XXXX binding as a general XXXX targeting mechanism. The influence of this work is two folds: It is the first experiment that directly XXXX, which greatly expand our understanding of how XXXX enters and infect the cell; It set up a technical framework for study other virus infectious pathway. Due to the scientific significance, this work has been highlighted in two recent journal review articles:

- Professor XXXX (XXXX, Institute of XXXX, XXXX) in his recent review “XXXX” [XXXX XXXX] summarized: “the best-studied XXXX”. In this review he highlighted our work many times while comparing the common and difference of entry mechanisms of two important members of the XXXX (mainly done by our works).
- XXXX (Principal Investigator, Institutes of XXXX, Centre for XXXX, University of XXXX) in his recent review “XXXX” [XXXX] cited my work when describe possible mechanism of XXXX.

(2) Grants: The results and the scientific importance of this project were central to a successful application of a \$ XXXX M funding support from XXXX.

Project 3: Year XXXX-XXXX, Ph.D. thesis research supervised by XXXX Professor XXXX XXXX Program and the Department of XXXX, University XXXX).

XXXX molecular mechanisms in live cells.

Another main project of my Ph.D. research is to study the XXXX mechanisms of XXXX in living cells. I utilized a novel quantitative fluorescent technique (called XXXX) to study the structure difference between inhibited and activated XXXX motors in live cells. Due to the sensitivity of this technique, new conformations of XXXX have been identified and novel molecular mechanisms that keep XXXX motor in an auto-inhibitory state have been discovered.

All proteins adapt different conformations in 3-D under various conditions. With no exceptions, XXXX motor changes conformations between inhibition and activation. The rough shapes of purified XXXX under different ion strength conditions have been estimated *in vitro* biochemically. However due to the sensitivity and resolution of the biochemical assay, detailed information about inhibited XXXX is not available. Furthermore, the true physiological conditions could not be replicated under these measurements. What conformations that XXXX adapts in living cells is totally unknown due to lack of experimental tools. To solve this problem, I decided to apply a newly developed quantitative fluorescence imaging method that is suitable to detect small distance differences at the 10 nanometer

range. This new method, named XXXX was originally developed to detect whether two probe-labeled proteins are getting close to each other within 10 nanometer distance in living cells. To use XXXX on one protein, I genetically modified XXXX to label both of the probes onto different positions of the same motor. Mapping distances between several pairs of labeled probes, I successfully obtained detailed spatial relationship (i.e. conformations) between different parts of XXXX. Comparing conformation changes between inhibited and active XXXX, novel auto-inhibition mechanisms of XXXX have been discovered.

These results were published in *Journal of XXXX*, one of the most highly rated journals in the XXXX field (XXXX. “XXXX.” XXXX). As of XXXX, Google Scholar search shows total of XXXX citations.

Significance and Impact:

(1) Being the first study of a XXXX motor protein structure and conformational change in living cells, my work has discovered new conformations of XXXX motor that would NOT be able to find by previous *in vitro* experiments. Here are some examples of other researchers citing my work in their publications:

- Dr. XXXX (XXXX) has cited my novel discovery that XXXX motor has multiple conformation as plausible explanation for his findings when study “XXXX” [XXXX *Journal*].
- Professor XXXX’s group (Department of XXXX, University XXXX) has used the XXXX I generated to study XXXX-mediated XXXX. Based on my work, they studied several XXXX structural variants and discovered that “XXXX” [XXXX].
- Followed by my pioneering discovery of the novel XXXX conformation *in vivo*, Professor XXXX’s group (Department of XXXX, XXXX University) continued working on refining this structural formation of XXXX with higher spatial resolution *in vitro* [XXXX].

(2) The results from this study led to a new model to explain the XXXX mechanism of XXXX, which is successfully tested by project 2 below. The following papers are examples of using this model as plausible mechanisms to explain their findings:

- Professor XXXX’s group (XXXX) has used the XXXX motor I generated to study “XXXX” [XXXX]. And appraised my model as example of diversified XXXX activation mechanisms.
- Professor XXXX (Department of XXXX) and Professor David Warshaw (Department of XXXX) in their recent review “XXXX” [XXXX] listed my work as “of outstanding interest” reference for the XXXX mechanism of motor regulation.

(3) The novel quantitative imaging technique XXXX I used in this work also opened a new window for study structure and conformational change of other proteins *in vivo*. Here are some examples of other researchers citing my work adapting FRET as principle technique in their publications:

- Professor XXXX’s group (Institute for XXXX) also used XXXX in his recent study about the association of two proteins, XXXX during XXXX entry of a cell [XXXX].

- Inspired by the XXXX study I have done Professor XXXX's group (Department of XXXX) used the same technique to study XXXX neck orientation in a structural perspective [XXXX].

(4) Due to the significance of this work, I was invited to give a platform presentation in XXXX Annual Meeting, XXXX XXXX). (The importance of the talk is similar to that of XXXX.)

Project 4: Year XXXX - XXXX, Ph.D. thesis research.

Discovered auto-inhibition and activation molecular mechanisms of XXXX in live cells and extended similar study to other XXXX motor protein family members previously less well studied.

At the same time, I am working on a sister project to study activation mechanism of XXXX in living cells. XXXX keeps itself as auto-inhibited in cell to prevent undesired ATP hydrolysis to save energy (see above). By what activation mechanism XXXX releases from auto-inhibited state before transport is totally unknown. Since I confirmed and discovered several important domains of XXXX that contribute to the auto-inhibition mechanism (see above), we hypothesized that molecules that interact with these domains may act as activation factors. In an *in vitro* yeast two-hybrid assay using two different inhibitory domains of XXXX as baits, my colleague XXXX successfully fished two protein molecules XXXX that bind to these two domains respectively. To reveal whether XXXX indeed play critical roles to activate XXXX in live cells, I developed a novel fluorescent imaging based assay (XXXX binding assay) and imaging analysis algorithms (calculating the motor protein relocation index). Using these novel techniques, I found that both XXXX are required for XXXX binding to microtubules in cells (set the trucks on highways). Adapting the single molecule assay I developed in the XXXX XXXX 1 Journal paper, I discovered that XXXX are not only required for XXXX binding but also processive movement on microtubule. A cooperative mechanism of XXXX "jump-start" is discovered.

I developed a novel fluorescent imaging based assay (called XXXX) and imaging analysis algorithms (calculating the motor protein relocation index) to study the XXXX activation mechanism and confirmed the function of two co-factors (identified by the first author of that paper in an *in vitro* screen) that coordinately activate XXXX in live cells.

From these two projects, molecular mechanisms of how XXXX molecule being kept in the auto-inhibition status (engine-shut-down in the parking lot) and being activated (jump start) are revealed. After these two projects, I guided another graduate student in the lab to study XXXX and XXXX motor proteins' biophysical properties and auto-inhibition mechanisms in living cells (see contribution 2 projects 3 described in achievement (1) above).

These results were published as a back-to-back article in the same issue of *Journal XXXX* (XXXX. In the same issue, these two papers are highlighted in an introductive short-article by Professor XXXX of XXXX University. As of XXXX, Google Scholar search shows total of XXXX citations.

Significance and Impact:

(1) Based on the projectproposed autoinhibition-activation mechanism, this study successfully identified XXXX as binding partners to the critical domains of XXXX. Using several novel imaging techniques I developed, XXXX are showed to co-activate XXXXmotor protein in live cells. These are the first co-

factors found for activating XXXX motor in mammalian cells. This research provided the first direct evidence for the model of activation mechanism proposed in project 1. XXXX play critical roles in the signaling pathways that control neuronal polarity and development and XXXX motor protein plays central roles in intracellular trafficking that tightly regulate the same neuronal process. Our results have been broadly cited not only in the motor protein but also in these research fields as key evidence previously missing to link various important elements in a complicated biological process.

- Being inspired by our study, Professor XXXX's group (Department of XXXX) has found the effector domain of XXXX as a novel binding partner that functions similarly to XXXX in our study to relieve XXXX another member of the XXXX family, XXXX [XXXX].
- Being inspired by our study, Professor XXXX's group (Department of XXXX) has reported "XXXX" [XXXX].
- Being inspired by our study showing the importance of XXXX controlled trafficking, Dr. XXXX's group (XXXX) has further study the structural implications of XXXX for cargo transport [XXXX].
- Professor XXXX's group (Faculty of XXXX) has found our result is supportive in their study of "XXXX" [XXXX].

Due to the importance of the imaging techniques and biological findings to the biophysics field, these two papers were used as core reading material in a course XXXX of the XXXX program of XXXX Institute of Technology taught by Professor XXXX (Department of XXXX).

(4) Grants: The results and the scientific importance of these two projects were critical for a successful application of a \$ XXXX funding support from XXXX.

Project 5: (XXXX-XXXX): Roles of XXXX

The history of life science teaches us that sometimes mechanisms concluded from *in vitro* data may not apply to real situation inside the cell due to the complexity of a biological system. Gathering direct evidences in live cells became the most crucial task for verifying the "road-sign" model even though supported by *in vitro* data. Furthermore, we want to know whether other members of the XXXX family, such as XXXX and XXXX motors follow similar rules when they face track choice problems. Taken this task, I developed some of the most state of art ultra-sensitive imaging techniques and combined with traditional cell biology and biochemical methods to study XXXX motor transport at the single molecule level in live cells. To find out correlation between various types of microtubules and XXXX motor transport, the key experiment was to trace XXXX motors and their cargoes overtime in cells with markers that show various microtubule properties. There were several technical difficulties that need to be solved to perform such experiment: (1) Ultra-sensitive imaging technique is required to trace as few as one XXXX motor undergoing active cargo transport in the cell. (2) To unambiguously differentiate signals from XXXX and Microtubule, they need to be labeled with spectrally distinguishable fluorophores and imaged in two separate channels. However, recording in separate channels result in challenge (3), aligning two images with nanometer precision.

As mentioned in project XX, I have developed XXXX microscope to image single XXXX motors in live cells. In this project, I applied the same idea to image XXXX and XXXX motor proteins in green channel. Differential labeling of microtubule markers can be achieved by using much red-shifted fluorophores and imaging in red channel. To assure absolute temporal synchronization between the two channels I optically separated the green-and-red mixing light and project each color image onto one half of the CCD camera. And finally I developed computer software to align the two channels corresponding to two half of CCD images at nanometer resolution. With all of these techniques developed, I am equipped to conquer the questions that could not be answer before. After laboratory testing of various combinations of different XXXX motors, cargoes and microtubule markers, I found that XXXX motor preferably select the stable microtubules that are chemically modified with acetylation and / or detyrocylation. Interestingly, XXXX motors avoid growing/unstable microtubules that are decorated by XXXX as if they read the “road-sign” saying “Detour – under construction”. Interestingly, XXXX and XXXX do not read the same “road-signs” as they unbiably transport on stable and growing microtubules. These observations helped explain why in the neuron, XXXX mostly transport to the axon that are filled with stable microtubules with acetylation and detyrocylation; while XXXX and XXXX transport to the dendrites that are more dynamic and lack of those two chemical modification. “Road-signs” that differentiate the status of a microtubule may be the key mechanism controls XXXX motor’s preferential transport in cells.

These results were published in XXXX, one of the most highly rated journals in the XXXX field of multidisciplinary (XXXX, XXXX citations).

Significance & Impact:

Combining the novel live cell single molecule imaging technique I developed (Project XX), I provided the first direct evidence that XXXX motor chooses stable microtubules, decorated with posttranslational modifications (PTM), over unstable/growing microtubules in cells. This important evidence answered a long-standing question about whether XXXX is a “smart” motor that can choose the right destination by itself. On the other hand, I found that XXXX and XXXX motors do not have the track preference like XXXX-1, which fits the more universal transport pattern of these two motors. The importance of microtubule PTM, and the differential transport mechanisms of different members of the XXXX family I discovered have inspired research of other labs and have been highlighted in review journals:

- My discovery of the importance of XXXX has inspired Professor XXXX’s group (Department of XXXX) to study the functionality of this XXXX in cilia; and found “XXXX” [XXXX].
- Inspired by my discovery that XXXX XXXX, Professor XXXX’s group (Department of XXXX) studied XXXX” [XXXX].
- Drs. XXXX (XXXX has highlighted my work in their recent review “XXXX” [XXXX]. In this review my discovery about XXXX was showed as important functions for “XXXX

My discovery about XXXX was included in one of the two most widely used XXXX textbooks in the world, XXXX). Two of my figures are put in a section on XXXX, which is edited by XXXX (Department of XXXX).

Due to my significant works in these two projects, I have been invited to give a platform presentation in the XXXX Annual meeting of the XXXX to present my important discovery of XXXX.

(5) I also co-authored a review in XXXX, one of the most highly rated review journals in the field of XXXX (XXXX; XXXX Google Scholar citations).

Publication list (total of XXXX Google Scholar citations as of XX/XX/XX):

- 1) XXXX (XXX citations)
- 2) XXXX (XX citations)
- 3) XXXX (featured on cover, XXX citations, top cited from XXXX to XXXX)
- 4) XXXX (XXX citations, media reports from XXXX)
- 5) XXXX (XX citations)
- 6) XXXX (XX citations)
- 7) XXXX (XX citations)